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H2**Started on** Tuesday, 25 August 2026, 5:08 PM**State** Finished**Completed on** Tuesday, 25 August 2026, 5:08 PM**Time taken** 31 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00**Problem Statement:**

Given a sorted array and a value x, the floor of x is the largest element in array smaller than or equal to x. Write divide and conquer algorithm to find floor of x.

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Value for x

Output Format

First Line Contains Integer – Floor value for x

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int findFloor(int arr[], int low, int high, int x) {
4      int ans = -1;
5
6      while (low <= high) {
7          int mid = low + (high - low) / 2;
8
9          if (arr[mid] <= x) {
10             ans = arr[mid]; // Potential floor, try to find a larger one on the right
11             low = mid + 1;
12         } else {
13             high = mid - 1; // Look on the left side
14         }
15     }
16
17     return ans;
18 }
19
20 int main() {
21     int n;
22     if (scanf("%d", &n) != 1) return 0;
23
24     int arr[n];
25     for (int i = 0; i < n; i++) {
26         scanf("%d", &arr[i]);
27     }
28
29     int x;
30     scanf("%d", &x);
31
32     printf("%d\n", findFloor(arr, 0, n - 1, x));
33     return 0;
34 }
```

	Input	Expected	Got	
✓	6 1 2 8 10 12 19 5	2	2	✓

	Input	Expected	Got	
✓	5	85	85	✓
	10			
	22			
	85			
	108			
	129			
	100			
✓	7	9	9	✓
	3			
	5			
	7			
	9			
	11			
	13			
	15			
	10			

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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